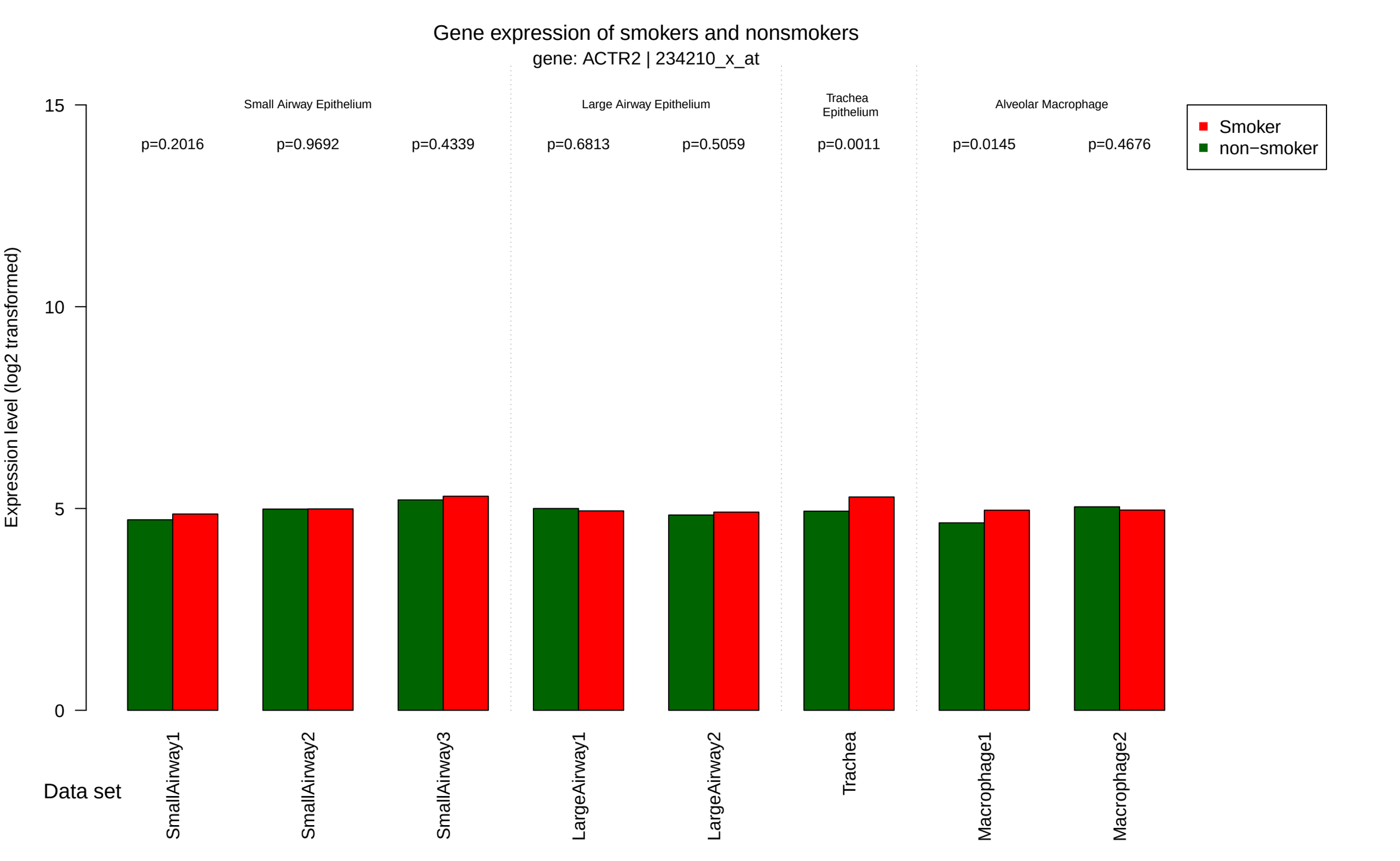
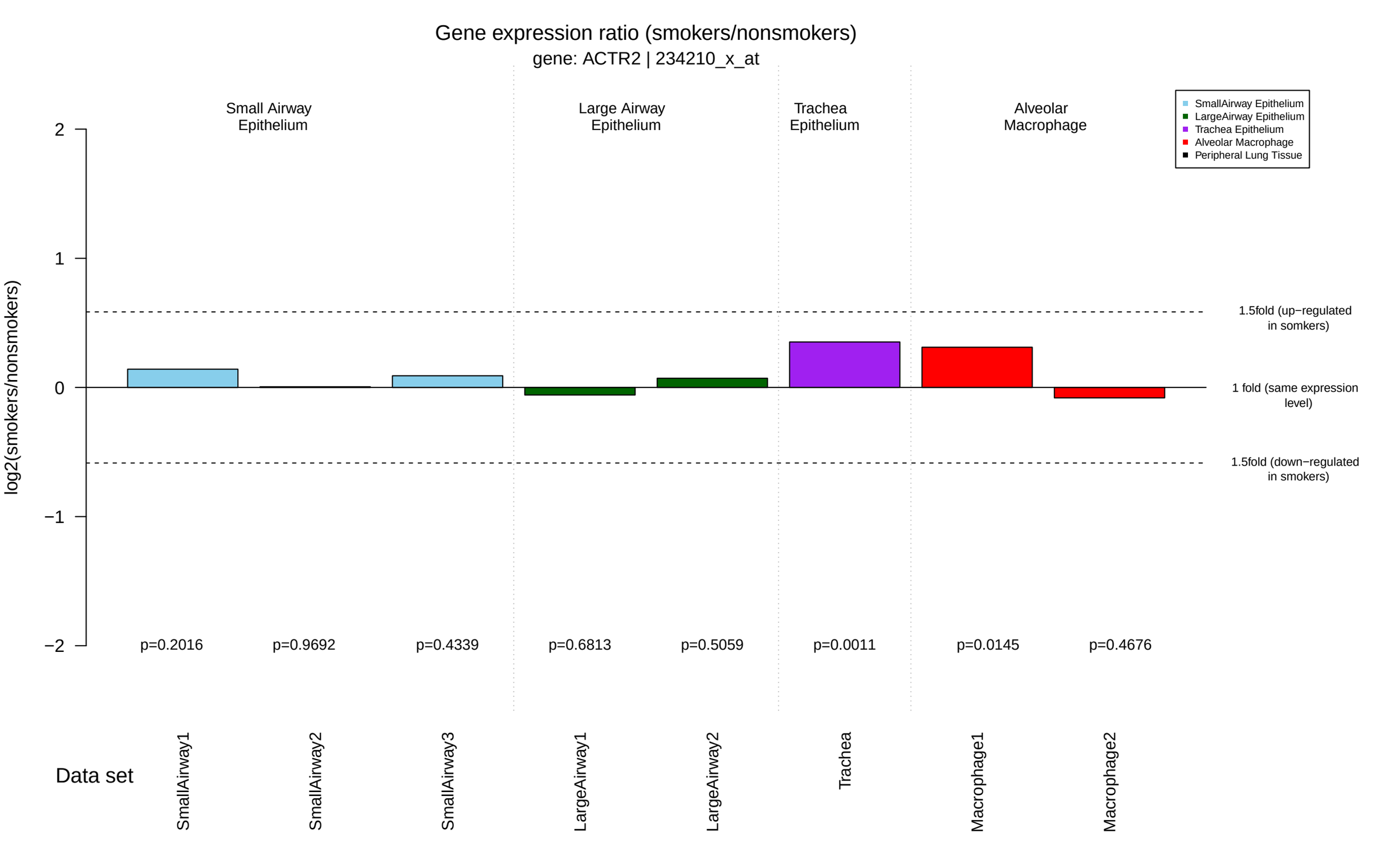
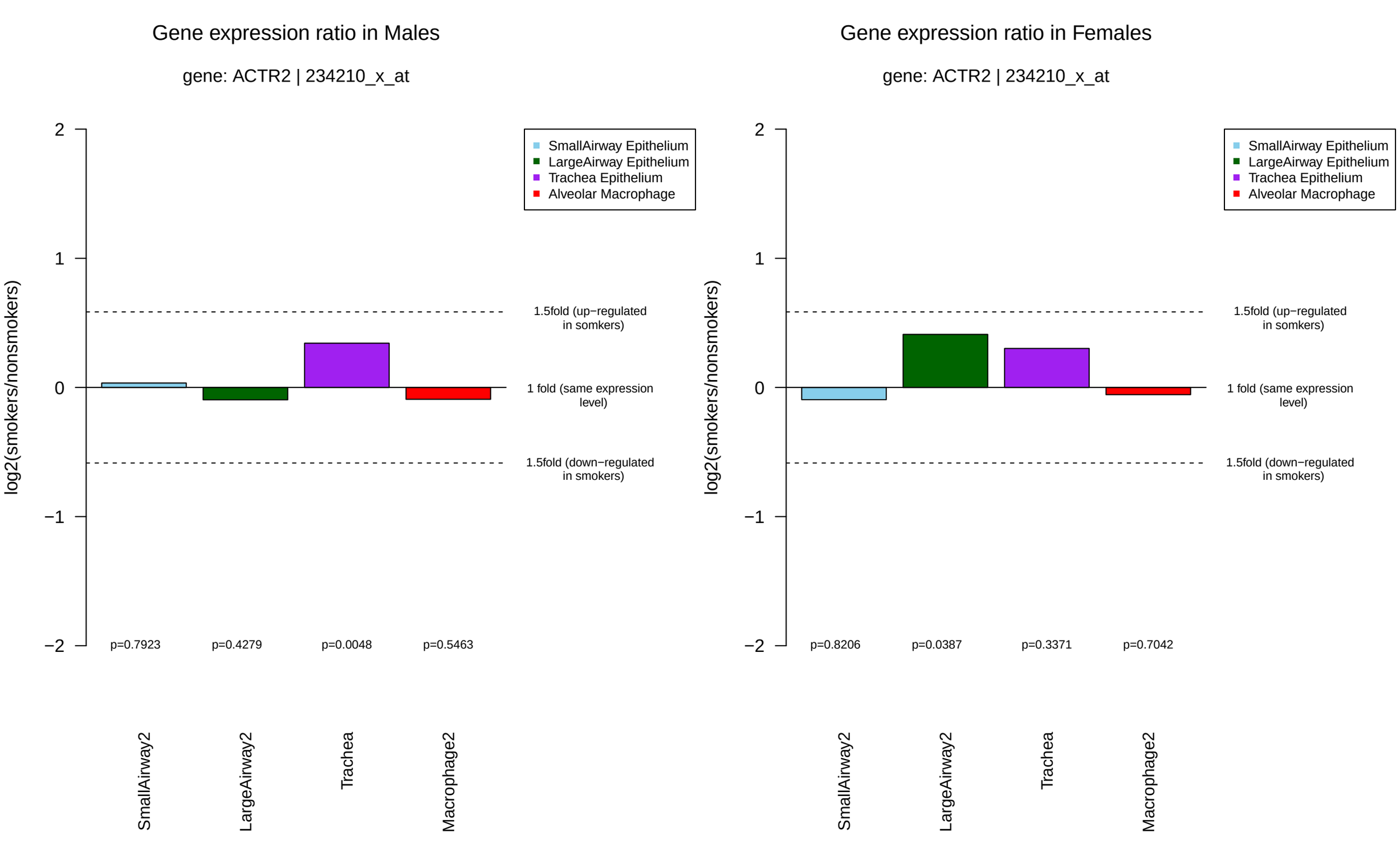
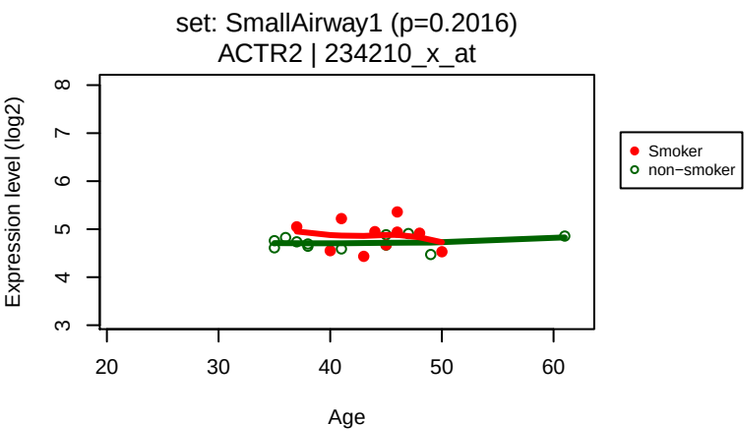
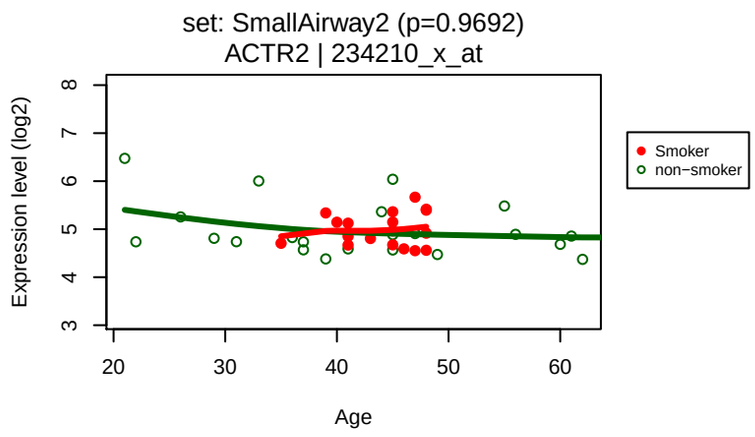
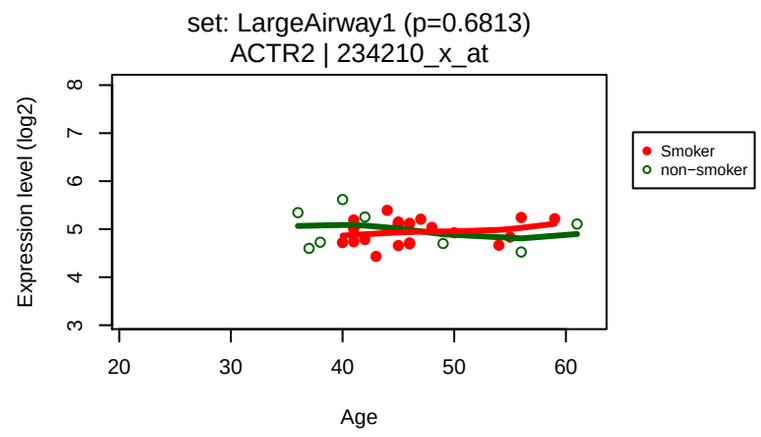

Gene expression of smokers and nonsmokers
gene: ACTR2 | 234210_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2016
p=0.9692
p=0.4339
p=0.6813
p=0.5059
p=0.0011
p=0.0145
p=0.4676
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

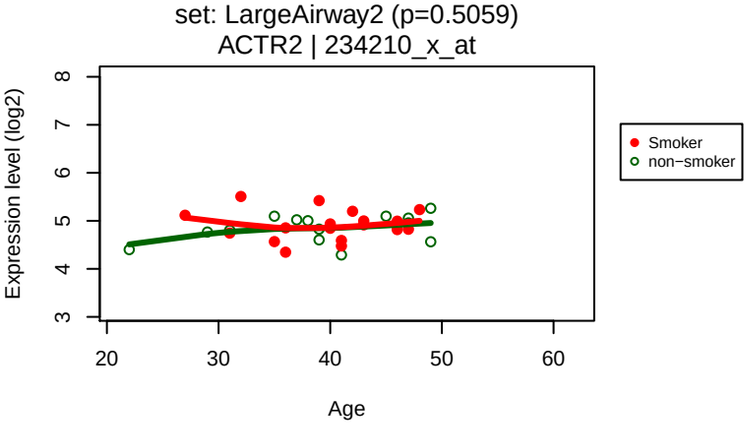

Gene expression ratio (smokers/nonsmokers)
gene: ACTR2 | 234210_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2016
p=0.9692
p=0.4339
p=0.6813
p=0.5059
p=0.0011
p=0.0145
p=0.4676
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

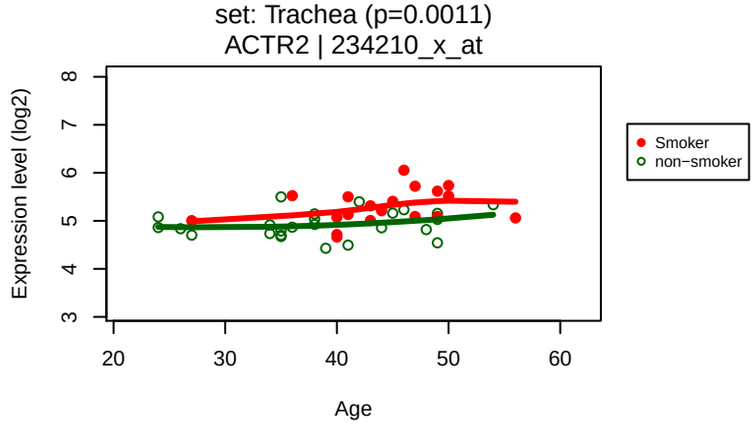

Gene expression ratio in Males
gene: ACTR2 | 234210_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7923
p=0.4279
p=0.0048
p=0.5463
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTR2 | 234210_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8206
p=0.0387
p=0.3371
p=0.7042
SmallAirway2
LargeAirway2
Trachea
Macrophage2

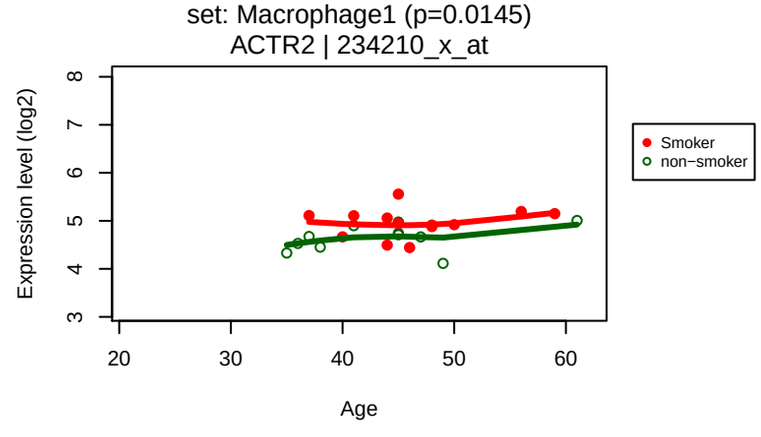

set: SmallAirway1 (p=0.2016)
ACTR2 | 234210_x_at
Expression level (log2)
Age
Smoker
non−smoker

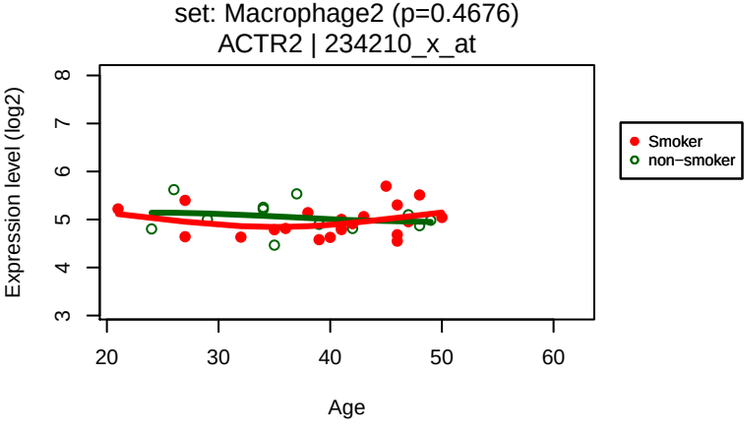

set: SmallAirway2 (p=0.9692)
ACTR2 | 234210_x_at
Expression level (log2)
Age
Smoker
non−smoker

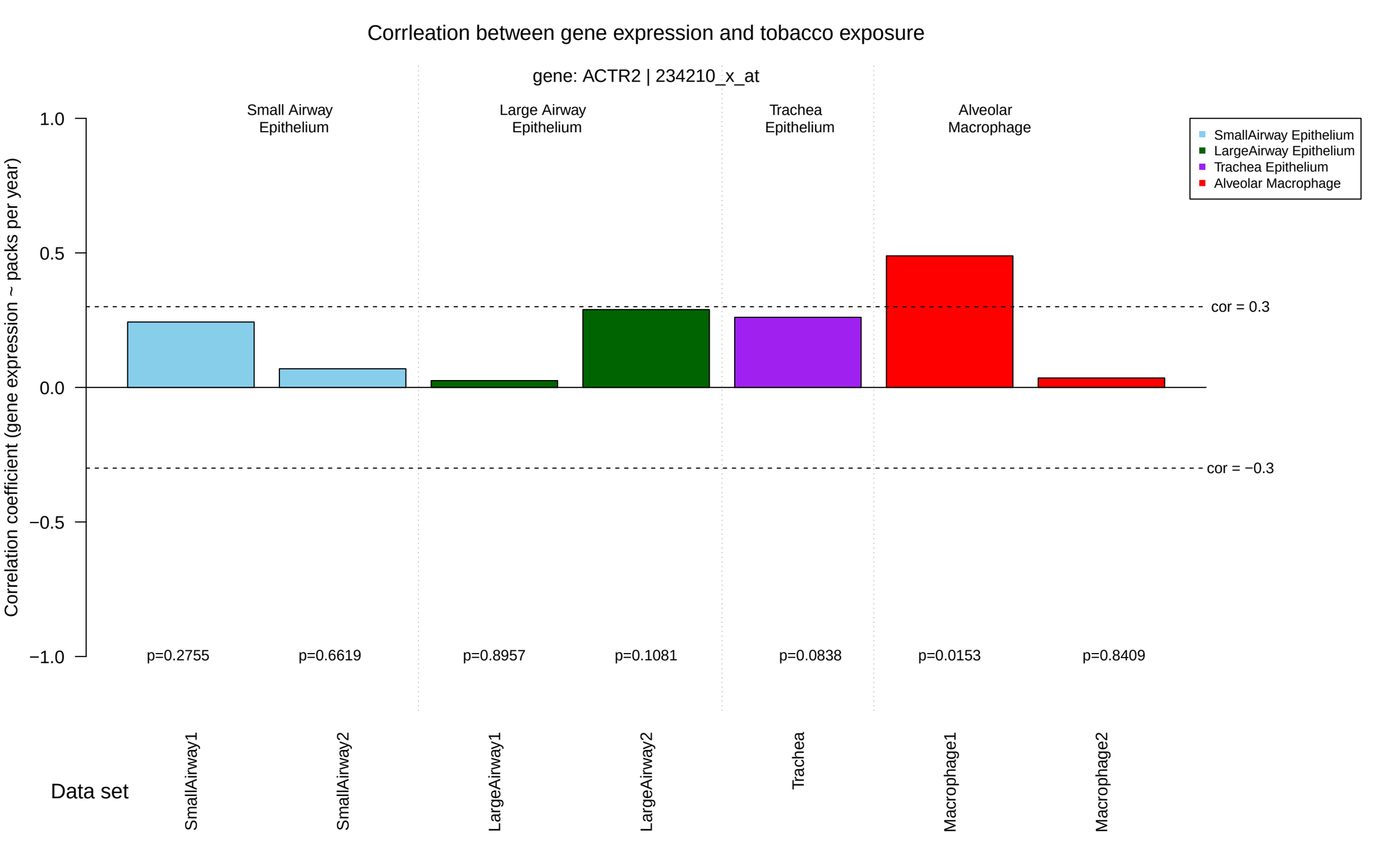

set: LargeAirway1 (p=0.6813)
ACTR2 | 234210_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5059)
ACTR2 | 234210_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0011)
ACTR2 | 234210_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0145)
ACTR2 | 234210_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4676)
ACTR2 | 234210_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTR2 | 234210_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2755
p=0.6619
p=0.8957
p=0.1081
p=0.0838
p=0.0153
p=0.8409
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2